

The Name of the Title Is Hope

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Abstract

LLM agents with web access are increasingly becoming the default for user-facing question answering. Reliable evaluation in this setting requires tests that remain unseen at train time and cannot be answered via verbatim lookup at inference. However, standard practice still relies on static benchmarks that age quickly and are vulnerable to contamination. We analyze two channels: pretraining leakage, where test items appear in the pretraining data, and retrieval-time leakage, where the system surfaces the exact target answer verbatim during evaluation and reports it rather than demonstrating genuine problem solving. We uncover evidence across five QA datasets for both channels and show that static evaluation can overestimate capability. We introduce DynamicWebQA, a dynamic evaluation framework for web-RAG that constructs questions and verifiable answers at evaluation time from current web sources. The framework enforces multi-document grounding, records evidence chains, and offers controllable complexity via source-set size, number of hops, and reasoning constraints. By generating fresh test instances, DynamicWebQA reduces memorization risk, mitigates retrieval-time leakage, and naturally covers time-sensitive queries.

CCS Concepts

• **Do Not Use This Code** → **Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

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1 Introduction

Benchmark leakage refers to the contamination that occurs when a model has prior exposure to a benchmark’s test items during pretraining, leading to inflated evaluation scores and undermining the benchmark’s validity [36, 37]. Since most LLMs are pretrained indiscriminately on publicly available web content, there is a high likelihood that they might have encountered benchmark datasets

hosted on platforms like HuggingFace, GitHub, Kaggle, and similar sites. If the base LLM used in a WebQA agent has already seen the test examples during pretraining, the evaluation effectively becomes a case of testing on the training set — severely compromising the credibility of the reported results.

Dynamic benchmarking is an emerging evaluation paradigm that generates or adapts test samples at evaluation time, so models are scored on fresh, unseen data and testset contamination is avoided [14, 20, 28, 35, 37, 38]. That means the agent does not see the same test sample twice, effectively addressing issues such as LLM memorization.

In this work, we refer to information that can change over time, such as the stock price of Tesla, as evolving information. Arguably, this type of content could be labeled as dynamic content or temporal content, but that might cause confusion, since we are already using the term “dynamic” to describe an evaluation paradigm (dynamic benchmarking), and “temporal” might lead readers to associate it with time-series data. To avoid such confusion, we adopt the term evolving information to specifically denote factual content that may shift over time without implying a particular format or evaluation framework.

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2 Related Work

Open-Domain QA (ODQA). Open-domain question answering studies how to find and use evidence from very large corpora to answer free-form questions. A sequence of datasets marked key shifts in task design: SQuAD [26] established span-based reading comprehension; Natural Questions [15] moved to real search queries; TriviaQA [12] stressed compositional, trivia-style questions with substantial lexical and syntactic variation; and HotpotQA [33] formalized multi-hop reasoning across multiple documents with sentence-level supporting facts and comparison questions, while including distractors to test *focus under noise*. Models such as fusion-in-decoder (FiD) [10] showed that aggregating many retrieved passages is a powerful way to synthesize scattered evidence.

Another important dimension in QA is *time*. RealTime QA [13] and FreshQA [31] evaluate whether systems can answer what is true *right now*, revealing failure modes when indices are stale or retrieved content is outdated or contradictory. These settings highlight needs that go beyond span extraction: recognizing when the answer has changed, when a question carries a false premise, and when recent evidence is required over older but higher-ranked documents. Finally, surface simplicity does not imply task simplicity. SimpleQA [32] demonstrates that short, seemingly straightforward questions can still require precise world knowledge or multi-step reasoning, indicating that difficulty is not well represented by length or phrasing.

Retrieval-Augmented Generation (RAG). Modern RAG approaches follow a common pipeline – given a query, retrieve top- k passages from a non-parametric memory and condition a generator on them. Foundational formulations include RAG [17], REALM [8], RETRO [2], and ATLAS [11], which together demonstrate that query-time retrieval improves knowledge-intensive tasks while enabling the external memory to be updated without re-training the parametric model. Recent surveys study components and variants of this paradigm [7].

In the literature, RAG is distinguished from broader tool-augmented generation – RAG requires *query-conditioned retrieval* whose contents directly condition the generator. Calculators, code execution, or static always-prepended context are typically treated as non-retrieval tools [21]. In practice, the retrieval index can be the open web or enterprise/local corpora, and the mechanism can be lexical (e.g., BM25) or dense; the generator then aggregates evidence across multiple passages [10, 17].

Beyond single-shot RAG, agentic frameworks let models iteratively issue searches, follow links, and cite sources. WebGPT augments an LLM with browsing and requires collecting references during answering [23]. ReAct further couples reasoning traces with tool calls (e.g., search, calculator) in a unified prompting loop, enabling multi-hop planning and verification [34]. These approaches often embed RAG steps inside a broader plan–act–observe loop, allowing complex queries that require cross-document synthesis.

Data Contamination. Data contamination arises when evaluation items (or close paraphrases and answer-bearing spans) leak into the corpora used for pretraining, instruction tuning, or fine-tuning. In such cases, test behaviour can devolve into *lookup* of memorized strings rather than the intended inference or problem

solving, inflating reported performance and obscuring brittleness [18, 27]. Prior studies connect leakage to memorization dynamics: larger models memorize more and earlier in training; duplication and longer prompts amplify verbatim regurgitation; and model capacity allows substantial unintended storage of datapoint-specific information [4, 22, 30]. At the same time, verbatim completion is not definitive evidence of membership—models can perfectly complete sequences never explicitly seen during training—so completion- or n -gram-based membership tests must be interpreted cautiously [19]. The measurable impact of contamination is further mediated by dataset construction, model scale, and properties of pretraining data [18].

Ravaut et al. [27] organize detection techniques by *data availability*. With candidate corpora available (*open-data*), methods compare evaluation items against sources via instance-/dataset-level span matching, embedding similarity, or paraphrase detection. When training data are unavailable (*closed-data*), researchers rely on indirect evidence such as performance analyses, membership-inference attacks, and instance-level probes of memorization and model confidence. Despite blind spots, approaches based on span matching remain the practical default in most foundational efforts [3, 5, 6, 16, 24, 25] – as they are auditable and more importantly, tractable at web scale. Auditing a single question or sentence against a large crawl is a needle-in-a-haystack search; doing so for entire benchmarks across billions of documents renders more sophisticated but compute-heavy detectors infeasible. Moreover, for closed models without training-set access, behavioural signals alone rarely certify contamination [18, 19].

Retrieval-augmented and agentic QA introduce a new concern – these systems may retrieve and quote target items from web or enterprise indices at *run time*, creating search-time data contamination that boosts scores without demonstrating reasoning [9]. Han et al. [9] show that even small leakage rates (e.g., on the order of a few percent) can reorder competitive leaderboards once online access is enabled, undermining trust relative to purely offline evaluation.

Dynamic Benchmarking. Most benchmarks for agentic or retrieval-augmented QA remain static and publicly accessible (e.g., HotpotQA), which makes them susceptible to contamination and memorization [4, 18, 22, 27, 33]. As models and training corpora grow, leakage can inflate reported capability and mask brittleness. This has motivated *dynamic* evaluation paradigms that resist memorization and better reflect real-world, evolving information needs. Early platforms like Dynabench and Dynaboard supported human-and-model-in-the-loop data collection and hosted live evaluations [14, 20, 29]. Other works instantiate dynamic settings in specific environments (e.g., AndroidWorld for interactive agents) [28] and in time-sensitive QA streams.

TBD

3 Modeling Contamination in RAG-Based QA

Retrieval-Augmented Generation (RAG) systems combine a parametric model M with external sources retrieved at inference. This creates two leakage channels: (i) *pretraining contamination*, where an evaluation item (or a close paraphrase / answer-bearing span) appears in the corpus used to pretrain M ; and (ii) *retrieval-time*

contamination, where the retriever surfaces such material during evaluation, enabling lookup rather than reasoning. Our experiments instantiate web-enabled RAG (“web-RAG”), but the formalism below applies to RAG broadly.

3.1 Definitions

Let M be pretrained on corpus D_M . Let D_{eval} contain items $x = (q, a)$ with question q and answer a . Let $\mathcal{R}(q)$ be the set of documents returned by the retriever for query q . Following Ravaut et al. [27], let f be a similarity function and τ a threshold; $\tau=1$ corresponds to *strict/verbatim* contamination.

We study both *question-only* and *question+answer* leakage, noting that even input-only (question-only) leakage can shift measured performance [18]. We consider two views of an item:

$$\pi_{\text{in}}(q, a) = q \quad (\text{question-only, a.k.a. input-only}),$$

$$\pi_{\text{in+lab}}(q, a) = q \parallel a \quad (\text{question+answer, a.k.a. input+label}).$$

Contamination criterion. For a channel $C \in \{\text{pre, ret}\}$ and view $v \in \{\text{in, in+lab}\}$, we say D_{eval} is contaminated if

$$\exists x \in D_{\text{eval}}, \exists d \in \mathcal{S}_C(x) \text{ s.t. } f(\pi_v(x), d) \geq \tau, \quad (1)$$

with source sets

$$\mathcal{S}_{\text{pre}}(x) = D_M, \quad \mathcal{S}_{\text{ret}}(x) = \mathcal{R}(q).$$

When evidence must be explicitly *answer-bearing*, we additionally require a predicate $g(a, d) \in \{0, 1\}$ (e.g., answer mention or high-overlap), and record positives of

$$\exists x, d \text{ s.t. } f(q, d) \geq \tau_q \wedge g(a, d) = 1. \quad (2)$$

3.2 Implementation details

We instantiate f using five auditable open-data detectors frequently used in LLM pre-training studies and consolidated in [27]:

- **Exact Match (EM)** [6].
- **8-gram words** (GPT-2 pre-training study) [25].
- **13-gram words** (GPT-3 pre-training study) [3].
- **50-character substring** (GPT-4 technical report) [24].
- **8-gram words with 70% threshold** (PaLM pre-training study) [5].

Span-based detectors scale to long documents, handle the length mismatch between short questions and web pages, and yield concrete evidence spans. While semantic detectors (embeddings / paraphrase) may increase recall, they are harder to audit and become infeasible across disparate lengths; we therefore adopt span-based methods as primary signals, consistent with cautions in [19] and common audit practice [27].

Pretraining channel (candidate retrieval). Full-membership testing against all datapoints in web-scale pretraining corpora is infeasible. Following the candidate-set approximation in open-data audits [18], we index the CommonCrawl C4 corpus in an Elastic-compatible service (AWS OpenSearch). For each $x = (q, a)$, we retrieve top- k candidates with BM25 (default $k=1000$) and apply the detectors between π_{in} or $\pi_{\text{in+lab}}$ and each candidate document. Any positive flag indicates potential pretraining contamination.

Retrieval-time channel (web-RAG). For each q , we query SearxNG [1] with the raw question string, pool results from the first three result pages across underlying engines, and define $\mathcal{R}(q)$ accordingly. We (i) compute contamination using (1) and, when needed, (2) over this pool, and (ii) use the same pool as retrieval context for answer generation. This isolates run-time leakage introduced by search from memorization in M .

Reporting. For each dataset and channel we provide: (a) question-only and question+answer contamination rates; (b) per-detector breakdowns; and (c) overlap between pretraining and retrieval-time hits.

4 Method

We propose a method that leverages *seed graphs* to anchor future question generation, avoiding reliance on static datapoints that are vulnerable to memorization or data contamination. Each evaluation round produces a fresh QA pair from every graph, ensuring that no fixed test items can be leaked in advance while maintaining semantic comparability across rounds through the shared anchors.

4.1 Seed Graphs

At evaluation round t , we maintain a set $\mathcal{S}_t = \{G_1^{(t)}, \dots, G_{n_t}^{(t)}\}$ of seed graphs. Each graph encodes the evidence trail (e.g., queries, retrieved documents, answer node) for a distinct information need. To support evolving information, graphs are constructed at evaluation time.

4.2 QA Generation

The generation function Gen takes a seed graph $G_i^{(t)}$, a random seed $r_{i,t} \sim \mathcal{R}$, and a fixed generation configuration θ to produce one QA pair:

$$(q_{i,t}, a_{i,t}) = \text{Gen}(G_i^{(t)}; r_{i,t}, \theta) \quad (3)$$

Here, θ denotes the fixed generation configuration (frozen prompt template and decoding settings, e.g., temperature, top- p , stop tokens). Gen is an LLM-based generator. $r_{i,t}$ is an i.i.d. sample from the randomness source $\mathcal{R}$, introducing per-sample stochasticity. Holding θ constant and varying only $r_{i,t}$ ensures that each system sees independent but identically distributed QA samples.

4.3 Sampling and Distributional Properties

Sampling an independent seed $r_{i,t} \sim \mathcal{R}$ for each graph yields the per-graph distribution:

$$\mathcal{P}_{i,t} = \text{Dist}(\text{Gen}(G_i^{(t)}; r, \theta)) \quad (4)$$

Because seeds are independent across graphs, the round- t dataset $D_t = \{(q_{1,t}, a_{1,t}), \dots, (q_{n_t,t}, a_{n_t,t})\}$ is an i.i.d. sample from the product measure:

$$\mathcal{P}_t = \prod_{i=1}^{n_t} \mathcal{P}_{i,t} \quad (5)$$

For a model M under evaluation, let $\text{score}((q, a), M)$ be any per-question metric (e.g., exact match, F1, log-loss). The aggregate dataset score is defined as:

$$\text{Agg}(D_t, M) = \frac{1}{n_t} \sum_{i=1}^{n_t} \text{score}((q_{i,t}, a_{i,t}), M) \quad (6)$$

By linearity of expectation and independence of $(q_{i,t}, a_{i,t}) \sim \mathcal{P}_{i,t}$:

$$\mathbb{E}_{D_t \sim \mathcal{P}_t} [\text{Agg}(D_t, M)] = \frac{1}{n_t} \sum_{i=1}^{n_t} \mathbb{E}_{(q,a) \sim \mathcal{P}_{i,t}} [\text{score}((q, a), M)] \quad (7)$$

This shows that the expected accuracy is simply the average of the per-graph expectations—so every seed graph contributes equally.

4.4 Collision Bound

Assume for every seed graph $G_i^{(t)}$, the generator can produce K distinct question–answer pairs:

$$C_{i,t} = \{(q, a) \in \mathcal{P}_{i,t} \mid \Pr((q, a)) > 0\}, \quad |C_{i,t}| = K$$

To characterize overlap across rounds $u < v$, define:

$$J_{i,u,v} = |C_{i,u} \cap C_{i,v}|, \quad J_{\max} = \max_{u < v} J_{i,u,v}$$

Because answers and evidence evolve over time, $J_{\max}$ tends to remain small for fast-changing topics.

Collision Probability. Sampling one QA pair per round, the probability that seed graph i produces the same QA pair twice within t rounds satisfies:

$$\Pr[\text{collision within } t] \leq \frac{t(t-1)}{2K^2} J_{\max} \quad (8)$$

Collision Control. To guarantee that this probability is at most $\delta \in (0, 1)$, it suffices to choose K such that:

$$K \geq \sqrt{\frac{t(t-1)}{2\delta} J_{\max}} \quad (9)$$

Equation (7) captures key trade-offs:

- Larger K (richer candidate pool) reduces collision probability for fixed t .
- Fewer evaluation rounds t allow a smaller K for the same error tolerance δ .
- Smaller $J_{\max}$ (i.e., greater semantic drift) reduces required K .

Because each seed graph is rebuilt from fresh evidence each round, its candidate set changes, helping keep $J_{\max}$ small. Equation (6) quantifies the collision risk, while Equation (7) provides a tunable design guideline for selecting K .

4.5 Cross-Round Reporting and Compatibility

Because each round rebuilds the seed graph set S_t , the underlying distribution $\mathcal{P}_t = \prod_i \mathcal{P}_{i,t}$ may drift over time, especially for evolving topics. Consequently, raw accuracies from different rounds may not be directly comparable. We offer two strategies:

i) **Snapshot Evaluation.** Fix a reference round t^* and freeze its seed graphs S_{t^*} . All systems are then evaluated on the same dataset D_{t^*} , ensuring identical test conditions.

ii) **Macro-Averaged Score.** For longitudinal tracking, aggregate performance across a window $\{t_1, \dots, t_T\}$ (e.g., most recent T rounds) using:

$$\text{Score}_{\text{macro}} = \frac{1}{T} \sum_{j=1}^T \text{Agg}(D_{t_j}, M) \quad (10)$$

Each D_{t_j} is i.i.d. from its own $\mathcal{P}_{t_j}$, so Eq. (8) summarizes average effectiveness under the benchmark’s natural evolution, without letting any single round dominate.

4.6 Verifying Distributional Consistency

If the generation is stable, items produced in different rounds should be indistinguishable once mapped into a semantics–preserving space. Sentence embeddings place short texts so that nearby points share topical content; consequently, a shift in topics should manifest as a change in the *distribution of pairwise distances*—not just a mean shift.

Representation. We concatenate each question–answer pair and compute a sentence embedding with `intfloat/e5-base-v2`. Vectors are ℓ_2 –normalized so Euclidean and cosine distances are monotone-equivalent.

Test protocol. On the Euclidean distance matrix we run a K –sample omnibus test with PERMANOVA (10k label permutations; report pseudo- F , p , and R^2). To localize potential differences, we run the Energy two-sample test for all $\binom{K}{2}$ round pairs (10k permutations; Benjamini–Hochberg FDR at $\alpha=0.05$). We add two descriptive visuals: a 2D PCA of embeddings colored by round, and a *centered mean-distance* map that removes within-round dispersion; this quantity is proportional to the biased energy distance and serves as an effect-size view.

5 Results

5.1 Contamination Modeling in QA Datasets

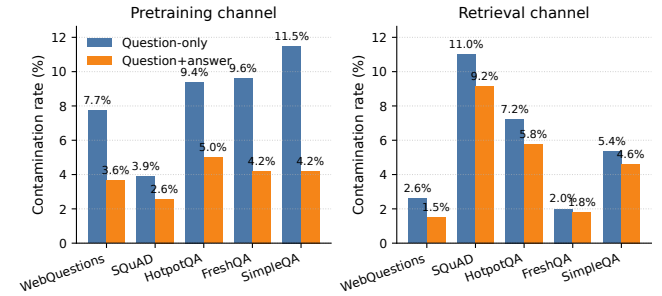


Figure 1: Contamination rates by dataset across the two leakage channels defined in the modeling framework: (a) pretraining contamination (source set $S_{\text{pre}}=D_M$), and (b) retrieval channel contamination (source set $S_{\text{ret}}=\mathcal{R}(q)$).

5.2 Model Performance

5.3 Distributional Consistency: Results

Omnibus. PERMANOVA detects no round effect in either split (Table 1). With 10,000 permutations the p –value resolution is 10^{-4} ; both splits yield $p \geq 0.99$ and $R^2 \leq 0.134\%$, indicating negligible variance explained by “round”.

Pairwise. For the 45 round pairs per split, no comparison is significant after BH–FDR ($\alpha=0.05$). E–statistics are tiny (10^{-4} – 10^{-3}) and centered near zero (Table 2); maxima are $< 10^{-3}$.

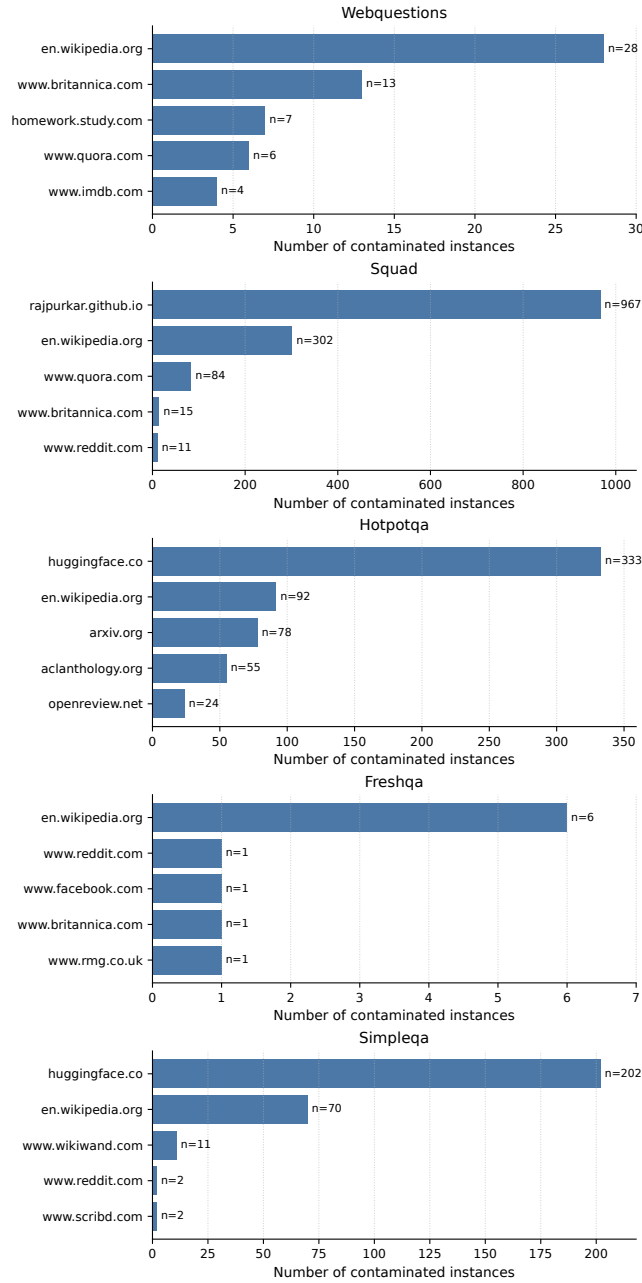


Figure 2: Top contamination sources during retrieval for each dataset. For each dataset, we show the top-5 domains by number of contaminated instances.

Visual evidence. PCA projections (Fig. 5) show full intermixing by round (PC1/PC2 <4% explained each). Drift heatmaps of signed E-statistics (Fig. 6) concentrate near zero without structure. Centered mean-distance maps and relative deviations (Fig. 7) show max absolute effects $\leq 1.5 \times 10^{-3}$, i.e., <0.25% of the within-round scale.

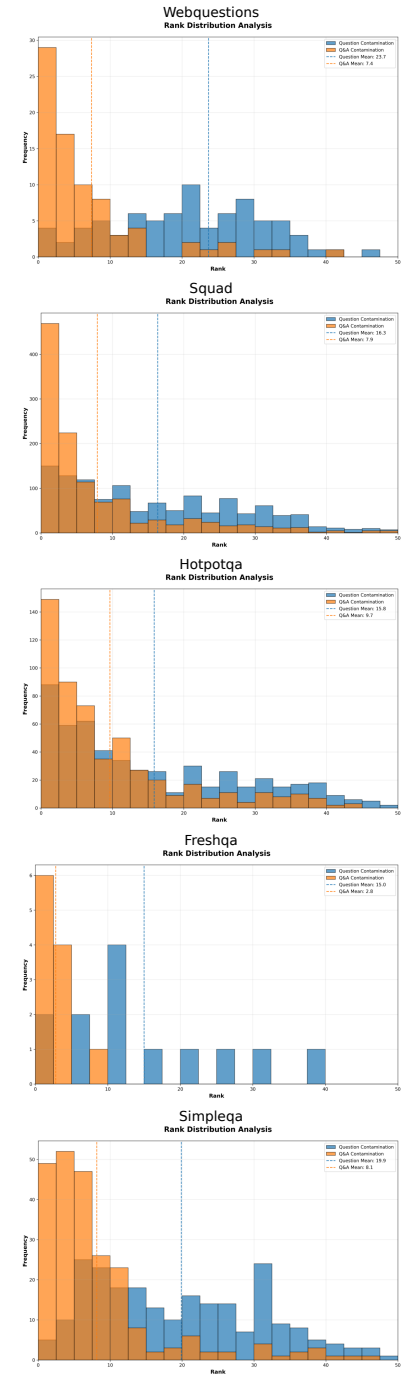


Figure 3: Rank distribution of contaminated evidence surfaced by the retriever, shown per dataset. Histograms separately show question-only and question+answer contamination where available.

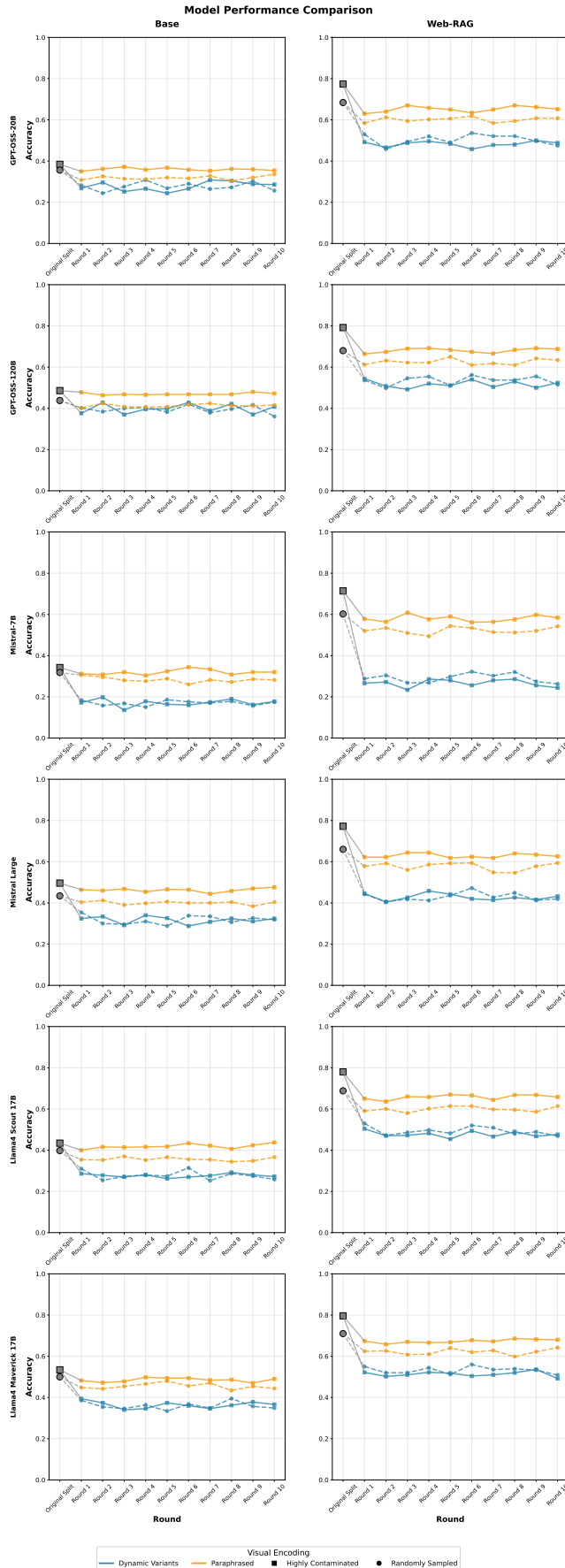


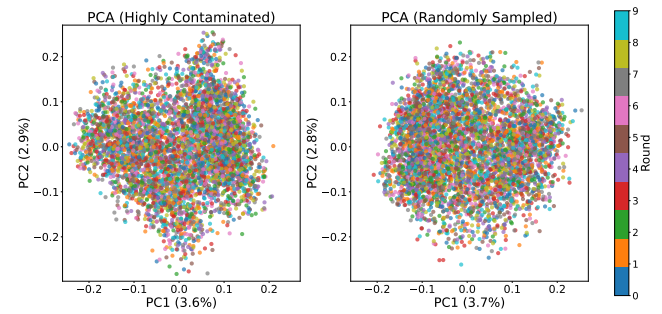
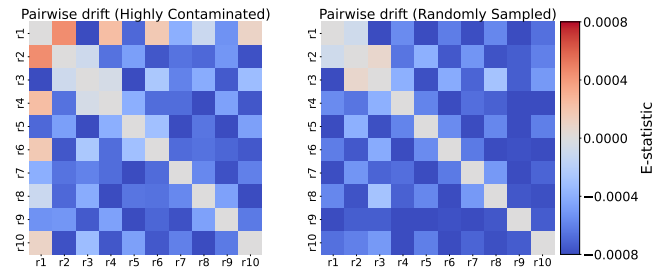
Figure 4: Caption TBD

Table 1: Omnibus PERMANOVA across $K=10$ rounds (Euclidean on ℓ_2 -normalized embeddings; 10k permutations).

Split	N	Pseudo- F	R^2 (%)	Perms	p -value
Highly Contaminated	5,000	0.7429	0.1338	10,000	≥ 0.99
Randomly Sampled	4,996	0.6476	0.1168	10,000	≥ 0.99

Table 2: Summary of pairwise Energy tests (signed, U-centered E-statistic).

Split	#Sig (FDR)	Mean	Median	Std
Highly Contaminated	0/45	-5.14×10^{-4}	-6.31×10^{-4}	3.16×10^{-4}
Randomly Sampled	0/45	-6.58×10^{-4}	-7.10×10^{-4}	2.08×10^{-4}

Figure 5: PCA projections of ℓ_2 -normalized embeddings for Highly Contaminated (left) and Randomly Sampled (right) splits. Points are colored by round; both splits show full intermixing with PC1/PC2 each explaining $<4\%$ variance.Figure 6: Pairwise drift heatmaps. Left: Highly Contaminated; Right: Randomly Sampled. Cells show the E -statistic from the Energy two-sample test (hyppo.ksample.Energy) computed on Euclidean distances of ℓ_2 -normalized embeddings; 10k permutations with BH-FDR over 45 round pairs.

6 Conclusion

TBD

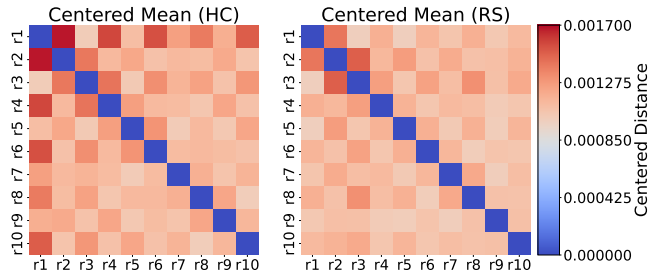


Figure 7: Centered mean distance across rounds. Left: Highly Contaminated; Right: Randomly Sampled. Each cell shows cross-round mean distance minus the average within-round distance (proportional to the biased Energy distance). Values cluster near zero with maxima $\leq 1.7 \times 10^{-3}$.

References

- [1] 2025. Searxng/Searxng. SearXNG.
- [2] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, Diego de Las Casas, Aurelia Guy, Jacob Menick, Roman Ring, Tom Hennigan, Saffron Huang, Loren Maggiore, Chris Jones, Albin Cassirer, Andy Brock, Michela Paganini, Geoffrey Irving, Oriol Vinyals, Simon Osindero, Karen Simonyan, Jack W. Rae, Erich Elsen, and Laurent Sifre. 2022. Improving Language Models by Retrieving from Trillions of Tokens. arXiv:2112.04426 [cs] doi:10.48550/arXiv.2112.04426 Comment: Fix incorrect reported numbers in Table 14.
- [3] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language Models Are Few-Shot Learners. arXiv:2005.14165 [cs] doi:10.48550/arXiv.2005.14165 Comment: 40+32 pages.
- [4] Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramer, and Chiyuan Zhang. 2023. Quantifying Memorization Across Neural Language Models. arXiv:2202.07646 [cs] doi:10.48550/arXiv.2202.07646
- [5] Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, Maarten Bosma, Gaurav Mishra, Adam Roberts, Paul Barham, Hyung Won Chung, Charles Sutton, Sebastian Gehrmann, Parker Schuh, Kensen Shi, Sasha Tsvyashchenko, Joshua Maynez, Abhishek Rao, Parker Barnes, Yi Tay, Noam Shazeer, Vinodkumar Prabhakaran, Emily Reif, Nan Du, Ben Hutchinson, Reiner Pope, James Bradbury, Jacob Austin, Michael Isard, Guy Gur-Ari, Pengcheng Yin, Toju Duke, Anselm Levskaya, Sanjay Ghemawat, Sunipa Dev, Henryk Michalewski, Xavier Garcia, Vedant Misra, Kevin Robinson, Liam Fedus, Denny Zhou, Daphne Ippolito, David Luan, Hyeontaek Lim, Barret Zoph, Alexander Spiridonov, Ryan Sepassi, David Dohan, Shivani Agrawal, Mark Omernick, Andrew M. Dai, Thanumalayan Sankaranarayanan Pillai, Marie Pellat, Aitor Lewkowycz, Erica Moreira, Rewon Child, Oleksandr Polozov, Katherine Lee, Zongwei Zhou, Xuezhi Wang, Brennan Saeta, Mark Diaz, Orhan Firat, Michele Catasta, Jason Wei, Kathy Meier-Hellstern, Douglas Eck, Jeff Dean, Slav Petrov, and Noah Fiedel. 2022. PaLM: Scaling Language Modeling with Pathways. arXiv:2204.02311 [cs] doi:10.48550/arXiv.2204.02311
- [6] Jesse Dodge, Maarten Sap, Ana Marasović, William Agnew, Gabriel Ilharco, Dirk Groeneveld, Margaret Mitchell, and Matt Gardner. 2021. Documenting Large Webtext Corpora: A Case Study on the Colossal Clean Crawled Corpus. arXiv:2104.08758 [cs] doi:10.48550/arXiv.2104.08758 Comment: EMNLP 2021 accepted paper camera ready version.
- [7] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2024. Retrieval-Augmented Generation for Large Language Models: A Survey. arXiv:2312.10997 [cs] doi:10.48550/arXiv.2312.10997 Comment: Ongoing Work.
- [8] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Mingwei Chang. 2020. Retrieval Augmented Language Model Pre-Training. In *Proceedings of the 37th International Conference on Machine Learning*. PMLR, 3929–3938.
- [9] Ziwon Han, Meher Mankikar, Julian Michael, and Zifan Wang. 2025. Search-Time Data Contamination. arXiv:2508.13180 [cs] doi:10.48550/arXiv.2508.13180
- [10] Gautier Izacard and Edouard Grave. 2021. Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering. arXiv:2007.01282 [cs] doi:10.48550/arXiv.2007.01282
- [11] Gautier Izacard, Patrick Lewis, Maria Lomeli, Lucas Hosseini, Fabio Petroni, Timo Schick, Jane Dwivedi-Yu, Armand Joulin, Sebastian Riedel, and Edouard Grave. 2022. Atlas: Few-shot Learning with Retrieval Augmented Language Models. arXiv:2208.03299 [cs] doi:10.48550/arXiv.2208.03299
- [12] Mandar Joshi, Eunsol Choi, Daniel S. Weld, and Luke Zettlemoyer. 2017. TriviaQA: A Large Scale Distantly Supervised Challenge Dataset for Reading Comprehension. arXiv:1705.03551 [cs] doi:10.48550/arXiv.1705.03551 Comment: Added references, fixed typos, minor baseline update.
- [13] Jungo Kasai, Keisuke Sakaguchi, Yoichi Takahashi, Ronan Le Bras, Akari Asai, Xinyan Yu, Dragomir Radev, Noah A. Smith, Yejin Choi, and Kentaro Inui. 2024. RealTime QA: What’s the Answer Right Now? arXiv:2207.13332 [cs] doi:10.48550/arXiv.2207.13332 Comment: RealTime QA Website: <https://realtimqa.github.io/>.
- [14] Douwe Kiela, Max Bartolo, Yixin Nie, Divyansh Kaushik, Atticus Geiger, Zhengxuan Wu, Bertie Vidgen, Grusha Prasad, Amanpreet Singh, Pratik Ringshia, Zhiyi Ma, Tristan Thrush, Sebastian Riedel, Zeerak Waseem, Pontus Stenetorp, Robin Jia, Mohit Bansal, Christopher Potts, and Adina Williams. 2021. Dynabench: Rethinking Benchmarking in NLP. arXiv:2104.14337 [cs] doi:10.48550/arXiv.2104.14337 Comment: NAACL 2021.
- [15] Tom Kwiatkowski, Jennimaria Palomaki, Olivia Redfield, Michael Collins, Ankur Parikh, Chris Alberti, Danielle Epstein, Illia Polosukhin, Jacob Devlin, Kenton Lee, Kristina Toutanova, Llion Jones, Matthew Kelcey, Ming-Wei Chang, Andrew M. Dai, Jakob Uszkoreit, Quoc Le, and Slav Petrov. 2019. Natural Questions: A Benchmark for Question Answering Research. *Transactions of the Association for Computational Linguistics* 7 (2019), 452–466. doi:10.1162/tacl_a_00276
- [16] Ariel N. Lee, Cole J. Hunter, and Nataniel Ruiz. 2024. Platypus: Quick, Cheap, and Powerful Refinement of LLMs. arXiv:2308.07317 [cs] doi:10.48550/arXiv.2308.07317 Comment: Workshop on Instruction Tuning and Instruction Following at NeurIPS 2023.
- [17] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *Advances in Neural Information Processing Systems*, Vol. 33. Curran Associates, Inc., 9459–9474.
- [18] Yucheng Li, Frank Guerin, and Chenghua Lin. 2024. An Open Source Data Contamination Report for Large Language Models. arXiv:2310.17589 [cs] doi:10.48550/arXiv.2310.17589
- [19] Ken Ziyu Liu, Christopher A. Choquette-Choo, Matthew Jagielski, Peter Kairouz, Sanmi Koyejo, Percy Liang, and Nicolas Papernot. 2025. Language Models May Verbatim Complete Text They Were Not Explicitly Trained On. arXiv:2503.17514 [cs] doi:10.48550/arXiv.2503.17514 Comment: Main text: 9 pages, 7 figures, 1 table. Appendix: 29 pages, 20 tables, 15 figures.
- [20] Zhiyi Ma, Kawin Ethayarajh, Tristan Thrush, Somya Jain, Ledell Wu, Robin Jia, Christopher Potts, Adina Williams, and Douwe Kiela. 2021. Dynaboard: An Evaluation-As-A-Service Platform for Holistic Next-Generation Benchmarking. arXiv:2106.06052 [cs] doi:10.48550/arXiv.2106.06052
- [21] Grégoire Mialon, Roberto Dessi, Maria Lomeli, Christoforos Nalmpantis, Ram Pasunuru, Roberta Raileanu, Baptiste Rozière, Timo Schick, Jane Dwivedi-Yu, Asli Celikyilmaz, Edouard Grave, Yann LeCun, and Thomas Scialom. 2023. Augmented Language Models: A Survey. arXiv:2302.07842 [cs] doi:10.48550/arXiv.2302.07842
- [22] John X. Morris, Chawin Sitawarin, Chuan Guo, Narine Kokhlikyan, G. Edward Suh, Alexander M. Rush, Kamalika Chaudhuri, and Saeed Mahloujifar. 2025. How Much Do Language Models Memorize? arXiv:2505.24832 [cs] doi:10.48550/arXiv.2505.24832
- [23] Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, Xu Jiang, Karl Cobbe, Tyna Eloundou, Gretchen Krueger, Kevin Button, Matthew Knight, Benjamin Chess, and John Schulman. 2022. WebGPT: Browser-assisted Question-Answering with Human Feedback. arXiv:2112.09332 [cs] doi:10.48550/arXiv.2112.09332 Comment: 32 pages.
- [24] OpenAI, Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altmenschmidt, Sam Altman, Shyamal Anadkat, Red Avila, Igor Babuschkin, Suchir Balaji, Valerie Balcom, Paul Baltescu, Haiming Bao, Mohammad Bavarian, Jeff Belgum, Irwan Bello, Jake Berdine, Gabriel Bernadett-Shapiro, Christopher Berner, Lenny Bognoffo, Oleg Boiko, Madelaine Boyd, Anna-Luisa Brakman, Greg Brockman, Tim Brooks, Miles Brundage, Kevin Button, Trevor Cai, Rosie Campbell, Andrew Cann, Brittany Carey, Chelsea Carlson, Rory Carmichael, Brooke Chan, Che Chang, Fotis Chantzis, Derek Chen, Sully Chen, Ruby Chen, Jason Chen, Mark Chen, Ben Chess, Chester Cho, Casey Chu, Hyung Won Chung, Dave Cummings, Jeremiah Currier, Yunxing Dai, Cory Decareaux, Thomas Degry, Noah Deutsch, Damien Deville, Arka Dhar, David Dohan, Steve Dowling, Sheila Dunning, Adrien Ecoffet, Atty Eleti, Tyna Eloundou, David Farhi, Liam Fedus, Niko Felix, Simón Posada Fishman, Juston Forte, Isabella Fulford, Leo Gao, Elie Georges, Christian Gibson, Vik Goel, Tarun Gogineni, Gabriel Goh, Rapha Gontijo-Lopes, Jonathan Gordon, Morgan Grafstein, Scott Gray, Ryan Greene, Joshua Gross, Shixiang Shane Gu, Yufei Guo, Chris Hallacy, Jesse Han, Jeff Harris, Yuchen He, Mike Heaton, Johannes Heidecke, Chris Hesse, Alan Hickey, Wade Hickey, Peter Hoeschele, Brandon Houghton, Kenny Hsu, Shengli Hu, Xin Hu, Joost Huizinga, Shantanu Jain, Shawn Jain, Joanne Jang, Angela Jiang, Roger Jiang, Haozhun Jin, Denny Jin, Shino Jomoto, Billie Jonn, Heewoo Jun, Tomer Kaftan, Lukasz Kaiser, Ali Kamali, Ingmar Kanitscheider, Nitish Shirish Keskar, Tabarak Khan, Logan Kilpatrick, Jong Wook Kim, Christina Kim, Yongjik Kim, Jan Hendrik Kirchner, Jamie Kiros, Matt Knight, Daniel Kokotajlo, Lukasz Kondraciuk, Andrew Kondrich, Aris Konstantinidis, Kyle Kopic, Gretchen Krueger, Vishal Kuo, Michael Lampe, Ikai Lan, Teddy Lee, Jan Leike, Jade Leung, Daniel Levy, Chak Ming Li, Rachel Lim, Molly Lin, Stephanie Lin, Mateusz Litwin, Theresa Lopez, Ryan Lowe, Patricia Lue, Anna Makanju, Kim Malfacini, Sam Manning, Todor Markov, Yaniv Markovski, Bianca Martin, Katie Mayer, Andrew Mayne, Bob McGrew, Scott Mayer McKinney, Christine McLeavey, Paul McMillan, Jake McNeil, David Medina, Aalok Mehta, Jacob Menick, Luke Metz, Andrey Mishchenko, Pamela Mishkin, Vinnie Monaco, Evan Morikawa, Daniel Mossing, Tong Mu, Mira Murati, Oleg Murk, David Mély, Ashvin Nair, Reiichiro Nakano, Rajeev Nayak, Arvind Neelakantan, Richard Ngo, Hyeonwoo Noh, Long Ouyang, Cullen O’Keefe, Jakub Pachocki, Alex Paino, Joe Palermo, Ashley Pantuliano, Giambattista Parascandolo, Joel Parish, Emy Parparita, Alex Passos, Mikhail Pavlov, Andrew Peng, Adam Perelman, Filipe de Avila Belbute Peres, Michael Petrov, Henrique Ponde de Oliveira Pinto, Michael, Pokorny, Michelle Pokrass, Vitchyr H. Pong, Tolly Powell, Alethea Power, Boris Power, Elizabeth Proehl, Raul Puri, Alec Radford, Jack Rae, Aditya Ramesh, Cameron Raymond, Francis Real, Kendra Rimbach, Carl Ross, Bob Rotsted, Henri Roussez, Nick Ryder, Mario Saltarelli, Ted Sanders, Shibani Santurkar, Girish Sastry, Heather Schmidt, David

- Schnurr, John Schulman, Daniel Selsam, Kyla Sheppard, Toki Sherbakov, Jessica Shieh, Sarah Shoker, Pranav Shyam, Szymon Sidor, Eric Sigler, Maddie Simens, Jordan Sitkin, Katarina Slama, Ian Sohl, Benjamin Sokolowsky, Yang Song, Natalie Staudacher, Felipe Petroski Such, Natalie Summers, Ilya Sutskever, Jie Tang, Nikolas Tezak, Madeleine B. Thompson, Phil Tillet, Amin Tootoonchian, Elizabeth Tseng, Preston Tuggle, Nick Turley, Jerry Tworek, Juan Felipe Cerón Uribe, Andrea Vallone, Arun Vijayvergiya, Chelsea Voss, Carroll Wainwright, Justin Jay Wang, Alvin Wang, Ben Wang, Jonathan Ward, Jason Wei, C. J. Weinmann, Akila Welihinda, Peter Welinder, Jiayi Weng, Lilian Weng, Matt Wiethoff, Dave Willner, Clemens Winter, Samuel Wolrich, Hannah Wong, Lauren Workman, Sherwin Wu, Jeff Wu, Michael Wu, Kai Xiao, Tao Xu, Sarah Yoo, Kevin Yu, Qiming Yuan, Wojciech Zaremba, Rowan Zellers, Chong Zhang, Marvin Zhang, Shengjia Zhao, Tianhao Zheng, Juntang Zhuang, William Zhuk, and Barret Zoph. 2024. GPT-4 Technical Report. arXiv:2303.08774 [cs] doi:10.48550/arXiv.2303.08774 Comment: 100 pages; updated authors list; fixed author names and added citation.
- [25] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. [n. d.]. Language Models Are Unsupervised Multitask Learners. (n. d.).
- [26] Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. 2016. SQuAD: 100,000+ Questions for Machine Comprehension of Text. arXiv:1606.05250 [cs] doi:10.48550/arXiv.1606.05250 Comment: To appear in Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP).
- [27] Mathieu Ravaut, Bosheng Ding, Fangkai Jiao, Hailin Chen, Xingxuan Li, Ruochen Zhao, Chengwei Qin, Caiming Xiong, and Shafiq Joty. 2025. A Comprehensive Survey of Contamination Detection Methods in Large Language Models. arXiv:2404.00699 [cs] doi:10.48550/arXiv.2404.00699 Comment: Accepted by TMLR in July 2025. 18 pages, 1 figure, 3 tables.
- [28] Christopher Rawles, Sarah Clincemaille, Yifan Chang, Jonathan Waltz, Gabrielle Lau, Marybeth Fair, Alice Li, William Bishop, Wei Li, Folawiyi Campbell-Ajala, Daniel Toyama, Robert Berry, Divya Tyamagundlu, Timothy Lillcrap, and Oriana Riva. 2025. AndroidWorld: A Dynamic Benchmarking Environment for Autonomous Agents. arXiv:2405.14573 [cs] doi:10.48550/arXiv.2405.14573
- [29] Tristan Thrush, Kushal Tirumala, Anmol Gupta, Max Bartolo, Pedro Rodriguez, Tariq Kane, William Gaviria Rojas, Peter Mattson, Adina Williams, and Douwe Kiela. 2022. Dynatask: A Framework for Creating Dynamic AI Benchmark Tasks. arXiv:2204.01906 [cs] doi:10.48550/arXiv.2204.01906 Comment: ACL System Demos 2022.
- [30] Kushal Tirumala, Aram H. Markosyan, Luke Zettlemoyer, and Armen Aghajanyan. 2022. Memorization Without Overfitting: Analyzing the Training Dynamics of Large Language Models. arXiv:2205.10770 [cs] doi:10.48550/arXiv.2205.10770
- [31] Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, and Thang Luong. 2023. FreshLLMs: Refreshing Large Language Models with Search Engine Augmentation. arXiv:2310.03214 [cs] doi:10.48550/arXiv.2310.03214 Comment: Preprint, 26 pages, 10 figures, 5 tables; Added FreshEval.
- [32] Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese, John Schulman, and William Fedus. 2024. Measuring Short-Form Factuality in Large Language Models. arXiv:2411.04368 [cs] doi:10.48550/arXiv.2411.04368 Comment: Blog post: <https://openai.com/index/introducing-simpleqa/>.
- [33] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. arXiv:1809.09600 [cs] doi:10.48550/arXiv.1809.09600 Comment: EMNLP 2018 long paper. The first three authors contribute equally. Data, code, and blog posts available at <https://hotpotqa.github.io/>.
- [34] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629 [cs] doi:10.48550/arXiv.2210.03629 Comment: v3 is the ICLR camera ready version with some typos fixed. Project site with code: <https://react-lm.github.io>.
- [35] Zhehao Zhang, Jiaao Chen, and Diyi Yang. 2024. DARG: Dynamic Evaluation of Large Language Models via Adaptive Reasoning Graph. arXiv:2406.17271 [cs] doi:10.48550/arXiv.2406.17271
- [36] Kun Zhou, Yutao Zhu, Zhipeng Chen, Wentong Chen, Wayne Xin Zhao, Xu Chen, Yankai Lin, Ji-Rong Wen, and Jiawei Han. 2023. Don't Make Your LLM an Evaluation Benchmark Cheater. arXiv:2311.01964 [cs] doi:10.48550/arXiv.2311.01964 Comment: 11 pages.
- [37] Kaijie Zhu, Jiaao Chen, Jindong Wang, Neil Zhenqiang Gong, Diyi Yang, and Xing Xie. 2024. DyVal: Dynamic Evaluation of Large Language Models for Reasoning Tasks. arXiv:2309.17167 [cs] doi:10.48550/arXiv.2309.17167 Comment: ICLR 2024 spotlight; 38 pages; code is at aka.ms/dyval.
- [38] Kaijie Zhu, Jindong Wang, Qinlin Zhao, Ruochen Xu, and Xing Xie. 2024. Dynamic Evaluation of Large Language Models by Meta Probing Agents. arXiv:2402.14865 [cs] doi:10.48550/arXiv.2402.14865 Comment: International

Conference on Machine Learning (ICML) 2024.

Appendix

Claim Extraction Prompt

You are given a passage of text. Your task is to extract one-sentence standalone factual claims from the passage.

Each claim must:

- Be self-contained (i.e., understandable on its own without requiring coreference resolution)
- Not rely on ambiguous references like “this”, “that”, “they”, etc.
- Be phrased as a complete declarative sentence
- Be verifiable (e.g., “Riot Games has no plans for League of Legends 2.” is acceptable; “They have no plans for League of Legends 2” is not)

For each claim, also provide the exact verbatim supporting text span from the original text that led you to extract the claim.

Output JSON Keys:

- "claim1": standalone claim
- "supporting_text_span1": verbatim supporting text span from original
- "claim2": standalone claim
- "supporting_text_span2": verbatim supporting text span from original
- ...

Task:

Here is the input text: {text}

Temporal-Sequence Multi-Hop QA Prompt

You are generating {n_pairs} **temporal-sequence** multi-hop QA pairs.

Buckets of claims (grouped by doc_id) are below:

{json.dumps(buckets, indent=2)}

Rules for each QA pair:

- Combine claims from at least {min_buckets} **different buckets**.
- Each chosen claim must mention a year, date, or explicit timeframe.
- Ask for an ordering or an interval (e.g., “How many years after ...?”).
- Every referenced fact must be essential for the answer.
- The question should not contain the answer or steps to get the answer.
- The question should not refer to the documents, they should be general.
- List every claim you used by its doc_id and claim_id.
- Provide one concise answer (year, number of years, earlier event, etc.).

Output format (JSON list):

- "used_claims": list of {{doc_id, claim_id, claim}}
- "question": string
- "answer": string

Return only the JSON—no extra commentary.

Comparison-Style Multi-Hop QA Prompt

You are generating $\{n_pairs\}$ **comparison-style** multi-hop QA pairs.

Buckets of claims (grouped by doc_id) are below:
`{json.dumps(buckets, indent=2)}`

Rules for each QA pair:

- Use claims from at least $\{min_buckets\}$ **different buckets**.
- Frame the question so the solver must contrast values (higher, lower, difference, ratio, earlier, later).
- Keep wording tight—only include values essential for the comparison.
- The question should not contain the answer or steps to get the answer.
- The question should not refer to the documents, they should be general.
- List every claim you used by its doc_id and claim_id.
- Provide a single, concise answer (number, entity, etc.).

Output format (JSON list):

- "used_claims": list of $\{\{doc_id, claim_id, claim\}\}$
- "question": string
- "answer": string

Return only the JSON—no extra commentary.

Conjunction-Style Multi-Hop QA Prompt

You are generating $\{n_pairs\}$ **conjunction-style** multi-hop QA pairs.

Buckets of claims (grouped by doc_id) are below:
`{json.dumps(buckets, indent=2)}`

Rules for each QA pair:

- Combine facts from at least $\{min_buckets\}$ **different buckets** (can use more).
- The answer must change if any referenced claim were false (logical AND).
- No trivia—every fact must be essential.
- The question should not contain the answer or steps to get the answer.
- The question should not refer to the documents, they should be general.
- List every claim you used by its doc_id and claim_id.
- Provide a single, concise answer (entity, number, date, etc.).

Output format (JSON list):

- "used_claims": list of $\{\{doc_id, claim_id, claim\}\}$
- "question": string
- "answer": string

Return only the JSON—no extra commentary.

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